

Task 1 - Install AD CS

- Open Server Manager.
- Click Manage
 - Add Roles and Features.
- Select Role-based or feature-based installation.
- Select your server.
- Check Active Directory Certificate Services.
- Click Add Features.
- Select Certification Authority.
- Click Next
 - Install.

Task 2 - Configure a Standalone Root CA

- After installation, click the notification flag in Server Manager.
- Click Configure Active Directory Certificate Services.
- Select Certification Authority.
- Choose Standalone CA.
- Select Root CA.
- Create a new private key.
- Keep the default cryptographic settings.
- Set the Common Name to:
 - CampusAppCA
- Complete the wizard.

Task 3 - Issue a User Certificate

If you don't already have the user:

- Open **Computer Management**
 - **Local Users and Groups.**
- Create a user named:
 - testuser

- Open Certification Authority.
- Right-click Issued Certificates (or use the certificate enrollment process).
- Request a certificate for testuser.
- After issuing, export the certificate.
- Export format:
 - testuser.cer

Task 4 - Simulate Secure Login

- Install Web Server (IIS) if needed.
- Create a simple test website.
- Open IIS Manager.
- Select your website.
- Open SSL Settings.
- Enable:
 - Require SSL
 - Require Client CertificatesConfigure Client Certificate Mapping Authentication.
- Map the certificate to testuser.
- Browse to the website using the issued certificate.

Task 5 - Troubleshooting Checklist

Common AD CS Problems

AD CS role not installed

- Verify Certification Authority role service is installed.

CA service stopped

- Open:
 - services.msc
- Restart:
 - Active Directory Certificate Services

Certificate templates unavailable

- Run:
 - certutil -catemplates

Client cannot request certificate

- Check:
 - DNS
 - Network connectivity
 - Permissions

IIS not accepting certificates

- Verify:
 - SSL enabled
 - Client certificates required
 - Correct certificate mapping

Intentional Misconfiguration

Example:

- Stop the CA service:
 - Stop-Service CertSvc

Try requesting a certificate (it should fail).

- Fix it:
 - Start-Service CertSvc
- Verify:
 - Get-Service CertSvc
- Status should be:
 - Running